WHAT HIDES MUSICAL GEOMETRY IN AUDIO REPRESENTATIONS?

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ABSTRACT

Learned representations are often evaluated by how well their attributes can be decoded, but decodability does not imply that a representation's ambient geometry reflects the relations among those attributes. Music makes the distinction testable, since relations between keys and between pitches are explicitly defined. Across nine audio representations, linear decoding, class-centroid geometry and readout geometry dissociate: a log-frequency spectrogram matches a self-supervised encoder at decoding key while showing no circle-of-fifths structure among its centroids, though its probe weights carry that structure strongly. Two factors govern how visible the geometry is: the number of examples behind each class centroid, which moves fifths geometry 13.4-fold where a thousandfold change in pooling duration moves it 1.7-fold, and the metric, under which octave equivalence absent from ambient distances is recovered by a rank-4 projection fitted on key labels alone. Helical pitch organization dissociates from octave equivalence in turn, the best helix fit belonging to the representation with no octave contrast.

***Index Terms*—** music representation learning, probing, representation geometry, key detection, octave equivalence

1. INTRODUCTION

Self-supervised music audio encoders support linear probes [1] for key, pitch, genre, and instrument [2, 3], and probe performance is commonly used to assess what information their representations make accessible. Yet successful decoding establishes accessibility, not the geometric form in which that information is organized. Two representations can support similarly accurate linear classifiers while arranging the corresponding classes differently in their embedding spaces. Music provides a direct setting for separating these questions because its attributes come with explicit relational structure. Musical keys are not merely 24 category labels: their relationships are organized by structures such as the circle of fifths and Krumhansl–Kessler profiles. Likewise, pitch contains both pitch height and pitch class, with octave equivalence relating notes separated by twelve semitones. These structures make it possible to ask not only whether key or pitch can be decoded, but whether distances within a representation reflect the expected relations between them.

We examine the same musical classes using three measurements, summarized in Table 1. Linear decoding measures how well class information can be recovered by a linear probe. We use class-centroid geometry to refer to the distances among class means in the original representation space, and probe-weight geometry to the distances among the corresponding classifier weights. Both are read by correlating those distances with a reference structure, as in representational similarity analysis [4].

Code, pre-registrations and all run artifacts: `https://github.com/Junst/topo-music`.

Table 1. The three measurements, all computed on the same clips, the same 24 keys and the same probe.

F1	**How easily key can be read out.** Macro-F1 of a multinomial logistic probe over 24 keys, folds grouped by track.
ρ_5^{cent}	**Whether the representation space is itself arranged by key relation.** Spearman correlation between the pairwise distances of the 24 key centroids and circle-of-fifths distance.
ρ_5^{W}	**Whether the directions the probe learns are arranged the same way.** The same correlation on the pairwise distances of its 24 class weight vectors.

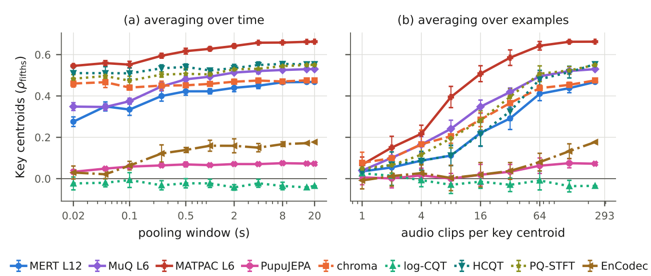

Fig. 1. Clip-level fifths geometry depends far more on the number of audio clips per centroid (b) than on temporal context (a). Error bars are one standard deviation over 5 and 10 draws.

Across nine audio representations, these measurements reveal distinct geometric profiles. We identify two factors that determine when musical geometry becomes visible. At the audio-clip level, circle-of-fifths geometry depends much more strongly on the number of examples used to estimate each key centroid than on temporal pooling duration. At the note level, octave equivalence depends strongly on the metric. Structure absent from ambient distances can remain weakly visible on a local manifold or become pronounced under a low-rank projection learned solely from key labels. We further show that helical pitch organization dissociates from octave equivalence, demonstrating that different geometric measurements of the same musical relation need not agree.

2. RELATED WORK

Probing Audio Representations. Linear probes are the standard instrument for asking what a representation encodes [1], and music audio encoders are routinely evaluated this way, with MERT [2] and MARBLE [3] reporting probe accuracy for key, pitch, genre and instrument. Castellon et al. [12] similarly evaluate music representations through shallow probes, including key detection. Hewitt and Liang [13] argue that probe performance should be interpreted rela-

Table 2. Key information, ambient geometry and readout geometry come apart, on the same 7035 GiantSteps clips and 24 keys. Blue marks the highest value in a column among the real representations and red the lowest. ρ_{chr} is the same correlation taken against semitone distance, a control on which a high value is a warning rather than a result.

representation	F1	ρ_5^{cent}	ρ_5^{W}	ρ_{chr}
Controls				
analytic fifths	0.181	+0.977	+0.986	−0.085
random features	0.039	+0.009	+0.033	−0.001
orthogonal key codes	0.613	−0.008	−0.005	−0.001
Spectral representations				
log-CQT [5], 12 bin/oct	0.392	**−0.035**	+0.560	**+0.174**
HCQT [6], 60 bin/oct	0.389	+0.554	+0.134	−0.038
PQ-STFT, 60 bin/oct	0.391	+0.550	+0.172	+0.112
chromagram [7]	0.346	+0.475	**+0.692**	+0.140
Neural codec				
EnCodec 32 kHz [8]	0.332	+0.177	+0.648	+0.041
MERT-v1-330M [2]				
default output, L24	0.340	+0.294	+0.484	+0.049
best layer, L4	0.379	+0.445	+0.412	−0.025
weighted sum, all layers	0.390	+0.514	+0.524	+0.149
MuQ-large [9]				
default output, L12	0.351	+0.372	+0.464	+0.056
best layer, L2	**0.418**	+0.533	**+0.103**	+0.021
weighted sum, all layers	0.418	+0.571	+0.412	−0.006
MATPAC++ music [10]				
default output, L12	0.329	+0.548	+0.656	−0.052
best layer, L6	0.390	**+0.663**	+0.366	**−0.061**
weighted sum, all layers	0.385	+0.656	+0.178	−0.056
PupuJEPA [11]				
teacher encoder	**0.198**	+0.072	+0.570	−0.007

tive to control tasks designed to measure probe selectivity, Voita and Titov [14] recast the measurement as description length rather than accuracy, and Belinkov [15] surveys what the instrument can and cannot support. A parallel line compares two representations to each other, as in centered kernel alignment [16], whereas we ask whether one representation matches a relational structure specified in advance.

Tonal Geometry. Music cognition has established well-defined relational structures for both key and pitch. Perceived key relationships follow the circle of fifths and Krumhansl–Kessler profiles [17, 18], while pitch perception separates pitch height from chroma [19]. Chroma features [5, 7] encode octave equivalence by construction and therefore provide a positive control for these relationships. Tonality-aware objectives such as STONE [20] explicitly impose key equivariance during training. Pilataki et al. [21] report a related observation in a JEPA-based transcription model, where NSynth embeddings show strong separation by timbre and pitch height but weak separation by pitch class. Closest to the octave measurements, Yagi et al. [22] fit a parametric helix to principal components of Jukebox [23] and MusicGen [24] representations of synthesized isolated notes, and report a helical arrangement whose clarity varies with the instrument and with the harmonic content of the signal.

3. METHOD

Representation decomposition. We decompose the representation z of a sample with class label y as

$$z = s_y + n_y + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid y] = 0, \tag{1}$$

where s_y is the class-dependent signal whose relations encode the structure of interest, n_y is other class-dependent variation and ε is the

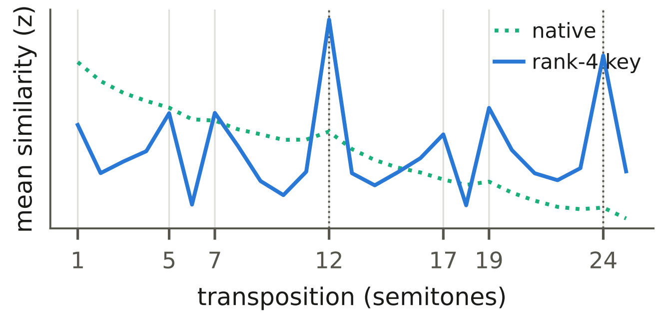

Fig. 2. A rank-4 projection fitted only on GiantSteps track tonics and applied unchanged to NSynth. The projected curve peaks at exactly twelve and twenty-four semitones, where the native one only falls off with distance.

residual. Averaging examples from the same class reduces ε but not n_y, whereas a change of metric can alter how s_y and n_y contribute to the measured geometry.

Representations. We consider three self-supervised music encoders, MERT-v1-330M [2], MuQ-large [9] and MATPAC++ [10], evaluated at matched relative depths. MuQ and MATPAC++ have half as many layers as MERT. We also include PupuJEPA [11], a music-domain JEPA over mel patches, using its teacher encoder, and EnCodec 32 kHz [8], a neural codec used by MusicGen [24]. The log-frequency representations comprise a log-CQT [5, 7] at 12 bins per octave, an HCQT [6] at 60 bins per octave stacked over six harmonics, and a multi-scale pitch-quantized STFT at 60 bins per octave. A CQT chromagram [7] folds octaves into 12 pitch classes and serves as a positive control. Frame-level representations are averaged over time to obtain one vector per clip.

Datasets. GiantSteps [25] contains 7,035 clips covering all 24 keys from 1,763 tracks, all within electronic dance music. FMAK [26] contains 5,476 usable tracks with expert key and mode annotations across 17 genres. We use one excerpt per track, preventing overlap across splits and reducing genre-specific confounding of key.

Key and readout geometry. In addition to the circle of fifths, centroid distances are compared with Krumhansl–Kessler key-profile distance and semitone distance (ρ_{chr}). The fifths and semitone references are nearly uncorrelated ($\rho = -0.038$). Significance is assessed with 2,000 permutations of the key labels. Partial Spearman correlation additionally controls for mel-spectral centroid distance. Probe folds are grouped by track.

Octave equivalence. On NSynth [27], we use only real recordings and exclude signal-processed pitch shifts. For each instrument, a fixed anchor set supports all transpositions up to 25 semitones. We measure the mean cosine similarity between each note and its transposition by k semitones. We report three preregistered statistics. The local contrast compares 12 semitones with the mean of 11 and 13 semitones, the strict contrast compares it with the maximum similarity from 8 to 11 semitones, and the periodic coefficient is obtained from a fitted twelve-semitone periodic term. Confidence intervals are bootstrapped over instruments, and all three statistics are required to agree in sign.

Metrics. We compute the octave statistics under three metrics on the same recordings. The native metric uses cosine distance in the original representation space. The geodesic metric uses shortest-path distance on a symmetric k-nearest-neighbor graph with cosine-weighted edges, as in Isomap [28]. We sweep k from 5 to 50 and report unreachable pairs without imputation. The supervised metric is induced by the linear projection described below. Distances are negated be-

Table 3. Centroid geometry, readout geometry and the chromatic control, measured on FMAK rather than GiantSteps: 5476 tracks over 17 genres, one excerpt per track. The three Δ columns are the NSynth octave contrast under the native metric and after a rank-4 projection fitted on GiantSteps and on FMAK keys. Δ^{nat} is measured on NSynth alone and so does not depend on the corpus.

representation	ρ_5^{cent}	ρ_5^W	ρ_{chr}	Δ^{nat}	Δ^{GS}	Δ^{FM}
Spectral representations						
log-CQT [5]	+0.057	+0.474	**−0.046**	+0.004	+0.466	+0.507
HCQT [6]	+0.389	+0.167	**−0.092**	+0.011	+0.535	+0.421
PQ-STFT	+0.454	+0.263	−0.055	**+0.227**	**+0.603**	**+0.513**
chromagram [7]	+0.632	+0.496	−0.092	+0.698	+0.878	+0.689
Neural codec						
EnCodec [8]	+0.207	+0.421	−0.076	+0.000	+0.459	−0.001
Learned encoders						
MERT L12 [2]	+0.437	+0.102	−0.068	−0.003	+0.223	+0.197
MuQ L6 [9]	+0.467	+0.184	−0.068	+0.034	+0.307	+0.204
MATPAC L6 [10]	+0.597	−0.064	−0.091	+0.010	+0.114	+0.069
PupuJEPA [11]	+0.031	+0.489	−0.078	**−0.004**	**+0.006**	**−0.002**

fore computing the contrasts, so positive values consistently indicate stronger octave equivalence.

Subspaces. We construct each supervised subspace from the top d Ledoit–Wolf shrinkage-LDA [29] discriminant directions for the specified target, with $d \leq 11$, the maximum rank for a twelve-class target. We report the fraction of total variance retained by each subspace. Matched random orthonormal subspaces control for the effect of dimensionality reduction itself. As a supervised null we repeat the same fit after permuting the GiantSteps key labels across tracks, which leaves the label distribution and the several-clips-per-track structure intact and destroys only the correspondence with audio, and report the resulting NSynth contrast over 200 draws.

Helical organization. We also fit the parametric pitch helix of Yagi et al. [22], which measures a different property of the same notes. For each instrument we average every pitch over a contiguous three-octave window, the span they use, and keep the 66 evaluation instruments that cover one. We apply PCA to those pitch means, enumerate the ten three-component projections of the top five components, fit the nine-parameter helix to each, and report the best. Their score is an inverse mean squared error and therefore depends on the scale of the embedding, so we report the fraction of pitch-centroid variance the helix explains, and read it against the same fit after the pitch order is shuffled, which leaves the point cloud intact and destroys only its correspondence with pitch.

4. RESULTS

4.1. Information, metric and readout dissociate

The controls in Table 2 establish that decodability alone does not imply fifths geometry. Orthogonal per-key codes achieve higher key F1 than any real representation while showing no fifths structure in either centroid or probe-weight geometry.

The log-CQT provides a direct example of this dissociation. It outperforms MERT layer 12 in grouped key decoding while showing no fifths structure in its centroids, despite strong fifths structure in its probe weights ($\rho_5^W = 0.560$). Randomly shuffled folds inflate performance for several representations and reverse the log-CQT–MERT comparison, motivating track-grouped evaluation throughout. The inflation is far from uniform, adding 0.33 to MERT layer 12 and to the pitch-quantized STFT but 0.03 to the chromagram, so the leak is largest where a representation can identify the track rather than the key.

Across depth, centroid and probe-weight geometry move in opposite directions in all three learned music encoders, while key F1 does not systematically improve. To control for post-hoc layer selection, we also follow MARBLE and learn a softmax-weighted combination of all hidden layers jointly with the probe. The weighted models match or slightly underperform the swept best layers when both are fitted with the same optimizer, with learned weights concentrating near the selected layers for MuQ and MATPAC.

The centroid–readout dissociations persist on FMAK, with no representation reversing its relative pattern. Chromatic proximity differs across corpora, appearing for several spectral representations on GiantSteps but not on FMAK.

4.2. Sample-limited masking

Fig. 1(a) shows relatively limited changes with pooling duration for most representations, with EnCodec as the main exception. By contrast, increasing the number of clips per centroid produces a much larger change. MERT L12 rises from 0.035 to 0.468, a 13.4-fold increase compared with 1.7-fold across the thousandfold pooling-duration sweep. With one example per centroid, fifths geometry is indistinguishable from zero for every representation.

This pattern is consistent with a dominant contribution from ε in Eq. (1), as within-key variation across tracks is reduced when more examples contribute to each centroid.

4.3. Octave equivalence depends on the metric

Under the native metric and the sign-consistency criterion registered in advance, MuQ shows a weak but reliable effect at every layer, whereas MERT and the codec largely fail it. Averaging NSynth recordings into pitch centroids recovers no reliable effect in MERT, so the near-zero geometry is not finite-sample variation alone. A k-nearest-neighbor geodesic metric reveals small but reliable effects in MERT layers 4 and 12 and in EnCodec, stable for k from 5 to 50, but not in layer 24. MERT layer 12 changes from -0.003 under the native metric to 0.012 under the geodesic one. Octave structure can therefore be absent from global distances, weakly preserved on the manifold, or visible only after supervised reweighting, and the geodesic metric is not uniformly more revealing.

The Δ^{nat} column of Table 3 shows that this metric dependence is not specific to learned representations, only the pitch-quantized STFT carrying substantial native octave structure.

Transfer of the supervised projection. The projection is fitted only to GiantSteps tonic labels, with no notes and no octave information, and applied unchanged to NSynth recordings. It raises the strict contrast to 0.346 for MuQ layer 2, and Fig. 2 shows the curve, with maxima at exactly twelve and twenty-four semitones. Refitting it on FMAK keys and applying it unchanged to the same recordings reproduces the transfer for MERT and the log-CQT, at 0.197 for MERT layer 12 against 0.223 from GiantSteps, even though MERT's clip-level readout geometry falls from 0.428 to 0.102 between the corpora, so the transfer does not depend on the corpus the projection was fitted on.

Alternative projections and nulls. Fig. 4 compares rank-4 subspaces fitted to different targets. Pitch-class supervision recovers octave equivalence for several representations, whereas pitch-height supervision does not show the same pattern. Permuting the key labels across tracks reduces the contrast to near the unfitted level, with none of 200 permutations reaching the true-label result for any learned

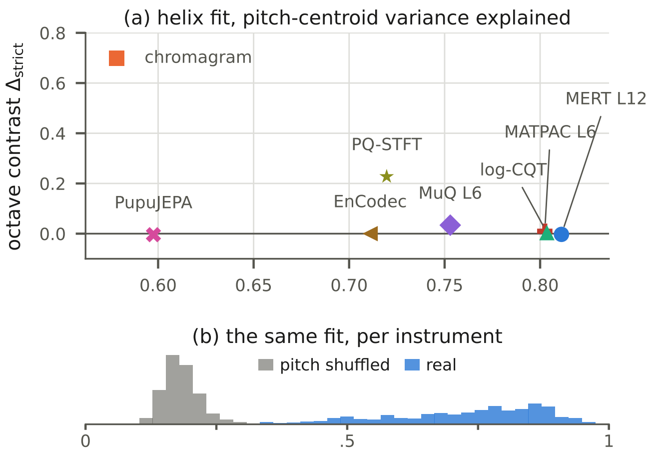

Fig. 3. Helical organization does not imply octave equivalence. (a) The two scores order the representations in opposite directions. (b) Every representation fits the pitch helix of Yagi et al. [22] far better than a pitch-shuffled reference, over 528 instrument fits.

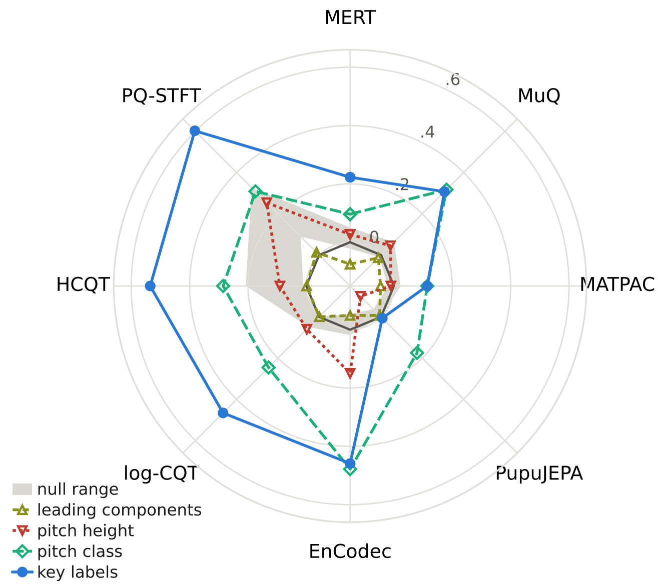

Fig. 4. The strict octave contrast on the same NSynth notes under rank-4 subspaces differing only in their fitting target. The band spans the two nulls, a random subspace and a permuted-label fit. Encoders are at MERT L12, MuQ L6 and MATPAC L6. The chromagram is omitted, a random rank-4 subspace of its twelve dimensions already reaching 0.569.

encoder. The key-supervised effect therefore depends on the correspondence between audio and key labels rather than on supervised low-rank projection alone. The permuted-label contrast is higher for the chromagram and pitch-quantized STFT, reaching 0.18 and 0.21, respectively, but remains below 0.04 for the learned encoders. The effect is also stable across projection rank for MERT layer 12, emerging at $d = 2$ and remaining between 0.21 and 0.24 through $d = 11$. PupuJEPA provides a contrasting case: its key-supervised contrast is near zero, while pitch-class supervision reaches 0.174, indicating that the null key-transfer result does not imply an absence of octave structure under other metrics. The learned subspaces retain less than 0.3% of the variance in NSynth.

Helical organization. Helicality measures a different geometric property from the preceding octave contrasts. A helix separates octaves along its height axis rather than identifying them. On the same notes, helical fit and octave equivalence rank the representations oppositely: MERT layer 12 has the strongest helix fit but weak octave equivalence, whereas the chromagram shows the reverse (Fig. 3).

Key supervision does not guarantee pitch-height invariance, since GiantSteps key labels may correlate with subgenre, tempo or production, though the transfer rules out an explanation resting only on GiantSteps-specific nuisance correlations. Nor is it mechanistic: whether the projection suppresses or amplifies is not identifiable here.

A failed registered prediction. We predicted that removing the recovered subspace, in the manner of nullspace projection [30] and amnesic probing [31], would at least halve clip-level fifths geometry. Against a baseline recomputed on the same held-out clips it barely moves, from 0.127 to 0.119 for MERT layer 12, and only the twelve-dimensional chromagram collapses. The subspace is therefore sufficient to expose the relation without exclusively containing it.

5. DISCUSSION

The results separate linear accessibility from the geometry in which musical classes are arranged. At the clip level, weak centroid geometry can result from limited class sampling, while at the note level the same musical relation changes substantially across ambient, geodesic, and supervised metrics. Probe accuracy and geometric similarity should therefore not be treated as interchangeable evidence that a representation has learned a musical relation. Geometric measurements depend on how examples are aggregated and how distances are defined, and comparisons should report these choices explicitly.

Limitations. NSynth is monophonic, our subspaces are linear by design, and the instrument split that separates fitting from evaluation on NSynth is a single draw. The FMAK-fitted projection does not transfer for EnCodec, where it falls from 0.459 to -0.001, so what a supervised projection recovers from a codec latent may itself be corpus-specific. Our helix numbers are also not directly comparable with those of Yagi et al. [22], who analyze different models on synthesized notes and score the fit by an inverse mean squared error that depends on the scale of the embedding. We report the scale-free form of the same fit and compare orderings rather than values.

6. CONCLUSION

We measured linear decodability, class-centroid geometry and readout geometry for the same musical classes, across nine audio representations and three corpora. The three come apart, and two factors account for most of the difference: at the clip level, how many examples estimate each centroid rather than how long each one is, and at the note level, the metric under which distances are compared. A rank-4 subspace fitted on clip-level key labels exposes octave periodicity in isolated notes that neither the ambient metric nor the leading principal components reveal, and neither a permuted-label fit nor a subspace fitted on pitch height reproduces it. Helical organization and octave equivalence order the same representations differently, so even two geometric readings of one relation need not agree. A musical relation therefore has no single geometric expression within a representation. Geometric claims should specify the measurement under which the structure appears.

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